THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapters 1–4: **Real Numbers, Polynomials, Pair of Linear Equations in Two Variables, Quadratic Equations** | Class 10 Mathematics | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 12 × 1 = 12

Q.	Correct Option	Brief Reasoning
1	**(a)** 4	$96 = 2^5 \times 3,\ 404 = 2^2 \times 101$; HCF $= 2^2 = 4$
2	**(c)** False, since $\sqrt{2} \times \sqrt{2} = 2$ is rational	Counter-example $\sqrt{2} \times \sqrt{2} = 2$
3	**(b)** Student A is correct	Co-prime ⇒ HCF $= 1$, LCM $= 8 \times 9 = 72$; B found the HCF
4	**(a)** 1	Number of distinct points on x-axis = number of zeroes = 1
5	**(b)** Incorrect — it can have at most 2 zeroes	A polynomial of degree n has at most n zeroes
6	**(d)** $3, -3$	$(x-3)(x+3)$
7	**(c)** $(3, 1)$	Adding: $2x = 6$
8	**(b)** $\frac{15}{4}$	$\frac{3}{2} = \frac{2k}{5} \Rightarrow k = \frac{15}{4}$
9	**(b)** $(3, 2)$	$2(3) + 3(2) = 12$ ✓
10	**(d)** $(x+1)^2 = x(x+3)$	(c) reduces to $x - 1 = 0$ (linear)
11	**(b)** $\frac{-b \pm \sqrt{b^2-4ac}}{2a}$	Quadratic formula
12	**(b)** Incorrect; $D = 0$, so the roots are equal $(3, 3)$	$D = 36 - 36 = 0$

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
13	$3825 = 5 \times 765 = 5 \times 5 \times 153 = 5^2 \times 9 \times 17 = 3^2 \times 5^2 \times 17$	2
14	$6^n = 2^n \times 3^n$. A number ending in 0 must have both 2 and 5 as prime factors; 5 is absent (uniqueness of prime factorisation). So 6^n never ends in 0.	1+1
15	$\alpha + \beta = 6,\ \alpha\beta = 8$; $\alpha^2 + \beta^2 = (\alpha+\beta)^2 - 2\alpha\beta = 36 - 16 = 20$	1+1
16	$x - y = 26,\ x = 3y \Rightarrow 2y = 26 \Rightarrow y = 13,\ x = 39$	1+1
17	$\sqrt{2}x^2 + 5x + 2x + 5\sqrt{2} = (x+\sqrt{2})(\sqrt{2}x+5) = 0 \Rightarrow x = -\sqrt{2}, -\frac{5}{\sqrt{2}}$. The student is **correct**.	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
18	Required number = HCF$(240, 1024)$. $240 = 2^4 \times 3 \times 5,\ 1024 = 2^{10} \Rightarrow$ HCF $= 2^4 = 16$.	1+1+1
19	$p(5) = 375 - 250 - 135 + 10 = 0$; $p(-2) = -24 - 40 + 54 + 10 = 0$; $p(\frac{1}{3}) = \frac{1}{9} - \frac{10}{9} - 9 + 10 = 0$ ✓. Sum $= \frac{10}{3} = -\frac{-10}{3}$ ✓	2+1
20	$5x + 7y = 50,\ 7x + 5y = 46$. Adding: $x + y = 8$; subtracting: $y - x = 2 \Rightarrow$ pencil ₹3, pen ₹5	1+1+1
21	$x(2x+1) = 528 \Rightarrow 2x^2 + x - 528 = 0$; $D = 4225 \Rightarrow x = \frac{-1+65}{4} = 16$. Breadth 16 m, length 33 m	1+1+1
22	$16 = 2^4,\ 20 = 2^2 \times 5$; LCM $= 2^4 \times 5 = 80$ min. They meet at **7:20 a.m.**	1+1+1
	OR: $616 = 2^3 \times 7 \times 11,\ 32 = 2^5$; HCF $= 2^3 = 8$. Maximum **8 columns**.	1+1+1
23	$3(x-1) = y,\ 4x = y + 8 \Rightarrow 4x = 3x + 5 \Rightarrow x = 5,\ y = 12$. Fraction $\frac{5}{12}$	1+1+1
	OR: $f - 2 = 5(s-2),\ f + 2 = 3(s+2) + 8 \Rightarrow 5s - 8 = 3s + 12 \Rightarrow s = 10,\ f = 42$ years	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
24	(i) 3 zeroes: $-2, 1, 3$ (ii) degree 3 (iii) $(x+2)(x-1)(x-3) = x^3 - 2x^2 - 5x + 6$ (iv) $-2 + 1 + 3 = 2 = -\frac{-2}{1}$ ✓	1+1+2+1
	OR: $(\alpha-\beta)^2 = (\alpha+\beta)^2 - 4\alpha\beta \Rightarrow 1 = 25 - 4k \Rightarrow k = 6$; zeroes 3 and 2 (3 marks). Zeroes $6, 4$: sum 10, product 24 ⇒ $x^2 - 10x + 24$ (2 marks).	3+2
25	$\frac{1}{x-3} + \frac{1}{x+5} = \frac{1}{3} \Rightarrow x^2 - 4x - 21 = 0 \Rightarrow (x-7)(x+3) = 0 \Rightarrow$ 7 years. Verify: $\frac{1}{4} + \frac{1}{12} = \frac{1}{3}$ ✓	2+2+1
	OR: $(x+2)(27-x) = 210 \Rightarrow x^2 - 25x + 156 = 0 \Rightarrow (x-12)(x-13) = 0 \Rightarrow$ (Maths, English) = (12, 18) or (13, 17)	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET B

Chapters 1–4: **Real Numbers, Polynomials, Pair of Linear Equations in Two Variables, Quadratic Equations** | Class 10 Mathematics | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 12 × 1 = 12

Q.	Correct Option	Brief Reasoning
1	**(c)** always irrational	If $r \cdot x = s$ (rational) then $x = s/r$ would be rational
2	**(a)** Student A is correct	$26 = 2 \times 13,\ 91 = 7 \times 13$; LCM $= 2 \times 7 \times 13 = 182$
3	**(b)** xy^2	Smallest powers: x^1y^2
4	**(b)** 4	$-\frac{-8}{2} = 4$
5	**(a)** 3	Degree 3 ⇒ at most 3 zeroes
6	**(d)** 5	$4 - 2k + 6 = 0 \Rightarrow k = 5$
7	**(d)** 10	$\frac{1}{5} = \frac{2}{k} \Rightarrow k = 10$; $\frac{3}{-7} \neq \frac{1}{5}$
8	**(b)** parallel to the y-axis	All points have $x = 4$
9	**(b)** no solution	Two distinct lines parallel to the x-axis
10	**(d)** $1, \frac{3}{2}$	$(2x - 3)(x - 1) = 0$
11	**(d)** 1	$4 + 2p - 6 = 0$
12	**(b)** 0	$D = 4 - 4(3)(\frac{1}{3}) = 0$

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
13	$12 = 2^2 \times 3,\ 15 = 3 \times 5,\ 21 = 3 \times 7$. HCF $= 3$; LCM $= 2^2 \times 3 \times 5 \times 7 = 420$	1+1
14	$\alpha + \beta = 5,\ \alpha\beta = 6 \Rightarrow \frac{\alpha+\beta}{\alpha\beta} = \frac{5}{6}$	1+1
15	$\frac{5}{7} \neq \frac{-4}{6}$ ⇒ unique solution ⇒ **consistent**	1+1
16	$9 - 6 - 15 = -12 \neq 0$, so 3 is not a root. $(x - 5)(x + 3) = 0 \Rightarrow 5, -3$	1+1
17	$4 - 4k = 0 \Rightarrow k = 1$; no real roots for $k > 1$	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
18	If $6 + \sqrt{2} = r$ (rational), then $\sqrt{2} = r - 6$ is rational — contradiction, since $\sqrt{2}$ is irrational.	1+1+1
19	Sum $= 6$, product $= 9 - 2 = 7 \Rightarrow x^2 - 6x + 7$. Check: $-\frac{b}{a} = 6,\ \frac{c}{a} = 7$ ✓	1+1+1
20	$j + 5 = 3(s + 5),\ j - 5 = 7(s - 5) \Rightarrow j = 3s + 10,\ 3s + 5 = 7s - 35 \Rightarrow s = 10,\ j = 40$ years	1+1+1
21	$l + b = 40,\ lb = 400 \Rightarrow x^2 - 40x + 400 = 0,\ D = 0 \Rightarrow x = 20$. Yes: $20\ \text{m} \times 20\ \text{m}$	1+1+1
22	Table: $x = -3, -2, -1, 0, 1, 2, 3, 4$ give $y = 6, 0, -4, -6, -6, -4, 0, 6$. The parabola cuts the x-axis at $(-2, 0)$ and $(3, 0)$ ⇒ zeroes -2 and 3.	2+1
	OR: $x^3 - (\text{sum})x^2 + (\Sigma\alpha\beta)x - \text{product} = x^3 - 2x^2 - 7x + 14$	3
23	$x^2 + (x + 2)^2 = 290 \Rightarrow x^2 + 2x - 143 = 0 \Rightarrow (x + 13)(x - 11) = 0$ ⇒ 11 and 13	1+1+1
	OR: $x^2 + (x + 30)^2 = (x + 60)^2 \Rightarrow x^2 - 60x - 2700 = 0 \Rightarrow (x - 90)(x + 30) = 0$ ⇒ 90 m and 120 m	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
24	(i) $48 = 2^4 \times 3,\ 72 = 2^3 \times 3^2,\ 108 = 2^2 \times 3^3$. (ii) LCM $= 2^4 \times 3^3 = 432$ s $= 7$ min 12 s ⇒ 7:07:12 a.m. (iii) $3600 \div 432 = 8.33$ ⇒ **8 times**.	1+2+2
	OR: Assume $\sqrt{7} = \frac{a}{b}$ (co-prime) ⇒ $7b^2 = a^2$ ⇒ $7 \mid a,\ a = 7c \Rightarrow b^2 = 7c^2$ ⇒ $7 \mid b$: contradiction. If $2 + 3\sqrt{7} = r$, then $\sqrt{7} = \frac{r-2}{3}$ is rational — contradiction.	3+2
25	$x + 20y = 1000,\ x + 26y = 1180 \Rightarrow 6y = 180,\ y = ₹30,\ x = ₹400$. For 30 days: $400 + 900 = ₹1300$	2+2+1
	OR: $(v + 10)(t - 2) = vt \Rightarrow -v + 5t = 10$; $(v - 10)(t + 3) = vt \Rightarrow 3v - 10t = 30 \Rightarrow t = 12$ h, $v = 50$ km/h; distance $= 600$ km	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET C

Chapters 1–4: **Real Numbers, Polynomials, Pair of Linear Equations in Two Variables, Quadratic Equations** | Class 10 Mathematics | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 12 × 1 = 12

Q.	Correct Option	Brief Reasoning
1	**(b)** $5 \times 7 \times 11 \times 13$	$5005 = 5 \times 1001 = 5 \times 7 \times 11 \times 13$
2	**(c)** 338	$26 \times 169 \div 13 = 338$
3	**(b)** 1	Consecutive numbers are co-prime
4	**(b)** $x^2 + 3x + 2$	$x^2 - (\text{sum})x + \text{product} = x^2 + 3x + 2$
5	**(d)** 4	$0x^5$ and $0x^3$ vanish; highest power with non-zero coefficient is 4
6	**(b)** Incorrect; zeroes are -2 and -5	$(x+2)(x+5) \Rightarrow -2, -5$
7	**(a)** infinitely many solutions	$\frac{1}{2} = \frac{2}{4} = \frac{3}{6} \Rightarrow$ coincident lines
8	**(c)** Incorrect — the pair has infinitely many solutions and is consistent	Coincident lines ⇒ dependent and consistent
9	**(c)** Incorrect — parallel lines give no solution, so the pair is inconsistent	Parallel lines ⇒ no common point
10	**(c)** ± 6	$k^2 - 36 = 0$
11	**(c)** no real roots	$D = 1 - 4 = -3 < 0$
12	**(a)** Student A is correct	$(x+3)(x-2) = 0$

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
13	$336 = 2^4 \times 3 \times 7$, $54 = 2 \times 3^3$. HCF $= 6$, LCM $= 2^4 \times 3^3 \times 7 = 3024$. $6 \times 3024 = 18144 = 336 \times 54$ ✓	1+1
14	$x^2 + 1 \geq 1 > 0$ for every real x, so it never becomes 0; its graph does not meet the x-axis. It has **no real zero** — the claim is wrong.	1+1
15	$\frac{1}{3} = \frac{2}{k} \Rightarrow k = 6$; then $\frac{c_1}{c_2} = \frac{-5}{15} = -\frac{1}{3} \neq \frac{1}{3}$ ✓. So $k = 6$	1+1
16	$3x + 3y = 15$; adding with 2nd: $5x = 19 \Rightarrow x = \frac{19}{5}$, $y = \frac{6}{5}$	1+1
17	$(10x - 1)^2 = 0 \Rightarrow x = \frac{1}{10}, \frac{1}{10}$	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
18	$11 \times 13 \times 17 + 17 = 17(143 + 1) = 17 \times 144$; $3 \times 5 \times 7 \times 11 + 11 = 11(105 + 1) = 11 \times 106$. Each has factors other than 1 and itself ⇒ composite.	1½+1½
19	$\alpha + \beta = \frac{5}{2}$, $\alpha\beta = \frac{3}{2}$. (i) $\frac{\alpha+\beta}{\alpha\beta} = \frac{5}{3}$ (ii) $\alpha\beta(\alpha+\beta) = \frac{15}{4}$ (iii) $(\alpha+\beta)^2 - 4\alpha\beta = \frac{25}{4} - 6 = \frac{1}{4}$	1+1+1
20	$x + y = 180$, $x - y = 18 \Rightarrow 99^\circ$ and 81°	1+1+1
21	$x(14 - x) = 50 \Rightarrow x^2 - 14x + 50 = 0$; $D = 196 - 200 < 0 \Rightarrow$ **not possible**	1+1+1
22	$5x - 5y = 100 \Rightarrow x - y = 20$; $x + y = 100 \Rightarrow$ 60 km/h and 40 km/h	1+1+1
	OR: $x + 4y = 27$, $x + 2y = 21 \Rightarrow y = ₹3$, $x = ₹15$	1+1+1
23	$18 = 2 \times 3^2$, $12 = 2^2 \times 3$; LCM $= 2^2 \times 3^2 = 36$. They meet again after **36 minutes**.	1+1+1
	OR: If $3\sqrt{2} = r$ then $\sqrt{2} = \frac{r}{3}$ is rational — contradiction. Hence $3\sqrt{2}$ is irrational.	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
24	$(3x - 4)(x + 1) \Rightarrow$ zeroes $\frac{4}{3}, -1$. Sum $\frac{1}{3} = -\frac{-1}{3}$ ✓, product $-\frac{4}{3}$ ✓ (3 marks). Reciprocals $\frac{3}{4}, -1$: sum $-\frac{1}{4}$, product $-\frac{3}{4} \Rightarrow 4x^2 + x - 3$ (2 marks).	3+2
	OR: (i) $-(x-1)(x-5) \Rightarrow x = 1, 5$ (ii) $5 - 1 = 4$ m (iii) sum $6 = -\frac{6}{-1}$, product $5 = \frac{-5}{-1}$ ✓ (iv) $-9 + 18 - 5 = 4$ m	2+1+1+1
25	Angle in a semicircle is 90°: $x^2 + (x+7)^2 = 169 \Rightarrow x^2 + 7x - 60 = 0 \Rightarrow (x+12)(x-5) = 0 \Rightarrow x = 5$. Yes; 5 m and 12 m from the gates	2+2+1

	OR: (i) $x(55-x)=750 \Rightarrow x^2-55x+750=0$ (ii) $(x-25)(x-30)=0$ ⇒ 25 or 30 toys (iii) Both 25 (₹30 each) and 30 (₹25 each) satisfy the condition; the claim is incomplete.	1+3+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND